

Supplementary Material

Inferring Institutionally Shaped Operational Constraints in Power System Dispatch Models

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S1 Forward UCED Formulation

The forward model solves independent 168-hour weekly unit-commitment and economic-dispatch (UCED) problems with fixed infrastructure. It co-optimizes unit commitment and dispatch, zonal exchange, storage operation, load shedding, and upward/downward operating reserves. The formulation below summarizes the principal constraint families and makes explicit where the inferred parameters enter the forward model.

Symbol	Meaning
$h \in H, i \in I, l \in L$	Hours, zones, and transmission corridors.
$G, G^{UC}, G^{CHP}, G^{VRE}$	Generators; commitment-based, CHP, and wind/solar subsets.
$s \in S, b \in B$	Electrical storage units and load-shedding segments.
$p_{g,h}, c_{s,h}, dc_{s,h}$	Generation, storage charging, and discharging (MW).
$o_{g,h}, su_{g,h}, sd_{g,h}$	Binary online, startup, and shutdown indicators.
$soc_{s,h}$	Stored energy (MWh).
$f_{l,h}^+, f_{l,h}^-, p_{i,b,h}^{LS}$	Directional corridor flows and unserved demand (MW).
$r_{a,h}^+, r_{a,h}^-$	Upward and downward reserve provision (MW).
$\xi_{l,h}^+, \xi_{l,h}^-$	Nonnegative MLT-band relaxation variables (MW).

Subscripts i on resource sets indicate location. For compactness, let $A = G \cup S$ denote reserve-providing resources and define $x_{a,h} = p_{a,h}$ for generators and $x_{a,h} = dc_{a,h}$ for storage. With one-hour time steps, the operating-cost objective is

$$\begin{aligned} \min \quad & \sum_{h \in H} \omega_h \left[\sum_{a \in A} VCG_{a,h} x_{a,h} + \sum_{i,b} VOLL_{i,b} p_{i,b,h}^{LS} \right] \\ & + \sum_{g \in G^{UC}, h} C_{g,h}^{SU} su_{g,h} + \pi^{MLT} \sum_{l,h} (\xi_{l,h}^+ + \xi_{l,h}^-), \end{aligned} \quad (S1)$$

where VCG , $VOLL$, and C^{SU} denote variable dispatch, involuntary load-shedding, and startup costs, respectively. In the implementation, $\omega_h = 1$ and $\pi^{MLT} = 10^7$.

Zonal power balance. For every zone and hour,

$$\sum_{g \in G_i} p_{g,h} + \sum_{s \in S_i} (dc_{s,h} - c_{s,h}) + \sum_l \Phi_{i,l,h} + \sum_b p_{i,b,h}^{LS} = Load_{i,h}, \quad (S2)$$

with

$$0 \leq \sum_b p_{i,b,h}^{LS} \leq Load_{i,h}. \quad (S3)$$

Here $\Phi_{i,l,h}$ denotes the loss-adjusted net injection from corridor l into zone i . Transmission losses are allocated symmetrically between the sending and receiving zones. Unless otherwise stated, continuous operating variables are nonnegative.

Generation output and unit commitment. For commitment-based generators,

$$AF_{g,h}\chi_{g,w}\mu_g P_g^{max} o_{g,h} \leq p_{g,h} \leq AF_{g,h} P_g^{max} o_{g,h}, \quad g \in G^{UC}, \quad (S4)$$

where $AF_{g,h}$ is hourly availability, μ_g is the minimum-output fraction, and $\chi_{g,w}$ is the seasonal CHP multiplier where applicable. For generators not subject to unit commitment,

$$0 \leq p_{g,h} \leq AF_{g,h} P_g^{max}, \quad g \in G \setminus G^{UC}. \quad (S5)$$

Commitment status satisfies

$$o_{g,h} - o_{g,h-1} = su_{g,h} - sd_{g,h}, \quad g \in G^{UC}, \quad (S6)$$

with $o_{g,h}, su_{g,h}, sd_{g,h} \in \{0, 1\}$. Minimum up- and down-time requirements are represented by

$$\sum_{t=h-UT_g+1}^h su_{g,t} \leq o_{g,h}, \quad (S7)$$

$$\sum_{t=h-DT_g+1}^h sd_{g,t} \leq 1 - o_{g,h}, \quad (S8)$$

with the corresponding boundary treatment applied at the beginning of each weekly solve. Commitment-adjusted ramp limits are imposed using the fixed unit-specific ramp-rate inputs RU_g and RD_g . Wind and solar generation are bounded by hourly availability, so curtailment is represented implicitly as available output minus dispatched output. Hydropower is represented using fixed availability and energy constraints in the implemented UCED; these constraints are omitted here because no hydropower parameter is varied in the inverse analysis.

Seasonal CHP requirement. During heating weeks, the committed CHP capacity satisfies

$$\sum_{g \in G^{CHP}} P_g^{max} o_{g,h} \geq 0.80 \sum_{g \in G^{CHP}} P_g^{max}. \quad (S9)$$

This is a heating-season commitment requirement rather than an output requirement. CHP output is additionally affected by the lower bound in Eq. (S4). In the case-study implementation, $\chi_{g,w} = 1.20$ in deep-winter weeks, 1.05 in the remaining heating weeks, and 1 otherwise; $\chi_{g,w} = 1$ for non-CHP generators.

Storage. Storage energy evolves according to

$$soc_{s,h} = soc_{s,h-1} + \eta_s^{ch} c_{s,h} - \frac{dc_{s,h}}{\eta_s^{dis}}, \quad 0 \leq soc_{s,h} \leq E_s^{max}. \quad (S10)$$

Storage is subject to standard intertemporal energy and shared power-capacity constraints, with fixed initial and terminal energy conditions. All storage parameters are held fixed in the inference.

Transmission and MLT schedules. Let ℓ_l denote the corridor loss fraction and $\kappa_l = 1 - \ell_l/2$. Directional flows satisfy

$$0 \leq f_{l,h}^+, f_{l,h}^- \leq \kappa_l F_l^{max}. \quad (S11)$$

For corridors with a signed MLT delivery benchmark $m_{l,h}$, the scheduled direction is determined by the sign of $m_{l,h}$ and the corresponding flow is constrained by

$$(1 - \delta)|m_{l,h}|\kappa_l - \xi_{l,h}^- \leq f_{l,h}^{dir} \leq (1 + \delta)|m_{l,h}|\kappa_l + \xi_{l,h}^+, \quad f_{l,h}^{opp} = 0. \quad (S12)$$

The parameter δ therefore controls the allowable delivery band around the scheduled MLT quantity. These schedule restrictions are imposed in the MLT-constrained formulation and omitted in the least-cost exchange reference case; physical corridor limits and designated one-way external exports remain in both formulations. The nonnegative slack variables soften the MLT bounds under a large penalty and are retained only as numerical/model diagnostics; no slack-based feasibility classifier or filtering rule is used in the inverse-inference workflow.

Zonal operating reserves. Resource-specific reserve-feasibility sets $\mathcal{R}_a$ impose power headroom, ramping, and, where applicable, storage-energy limits:

$$(r_{a,h}^{\uparrow}, r_{a,h}^{\downarrow}) \in \mathcal{R}_a(x, o, c, soc). \quad (\text{S13})$$

For every zone and hour,

$$\sum_{a \in A_i} r_{a,h}^{\uparrow} \geq \lambda_i^D \text{Load}_{i,h} + \lambda_i^{VRE} \sum_{g \in G_i \cap G^{VRE}} p_{g,h} + Q_i, \quad (\text{S14})$$

$$\sum_{a \in A_i} r_{a,h}^{\downarrow} \geq \lambda_i^D \text{Load}_{i,h} + \lambda_i^{VRE} \sum_{g \in G_i \cap G^{VRE}} p_{g,h}, \quad (\text{S15})$$

where Q_i is the configured largest-unit/interconnection contingency. Reserve coefficients and resource-level capability parameters are fixed within each study year.

The inverse-inference vector is

$$\theta = (\delta, \{\mu_k, \tau_k\}_{k \in K}), \quad (\text{S16})$$

where δ enters the MLT delivery band, μ_k enters the minimum-output bounds for units in class k , and the duration parameter enters through

$$UT_g = DT_g = \text{round}(\tau_k), \quad g \in k. \quad (\text{S17})$$

All remaining UCED inputs and constraint parameters are held fixed within each study year's inference experiment.

S2 Candidate Parameterization and Preliminary Screening

S2.1 Preliminary coal-side screening

A preliminary three-month screening exercise compared the responses of CHP and non-CHP coal parameters. CHP minimum-output parameters showed clearer coal-generation responses, particularly in the two smaller unit-size classes. Non-CHP parameters showed limited response under that design and were not carried into the full-year inference experiment. This is a design-specific screening result, not a claim that non-CHP flexibility is generally unimportant or intrinsically unrecoverable.

S2.2 Candidate domain and generator grouping

The full-year inference design uses a 100-point LHS design over a prespecified 13-dimensional domain: six minimum-output fractions, six common minimum up/down-time controls, and one MLT band. Table S1 lists the fixed domain boundaries, not realized sample minima and maxima.

R denotes official retrofitted coal pilot units, including both CHP and conventional coal where present. **C** denotes non-retrofitted CHP. These groups are crossed with three capacity classes in MW. The time control is sampled numerically and rounded to an integer before being assigned to both minimum up and down times. Other non-retrofitted conventional coal units use fixed minimum output 0.40 and up/down-time inputs of 8 hours in the input-generation workflow used for the reported experiments.

Table S1: Complete candidate domain. “min” is a capacity fraction; “time” is in hours. All 13 dimensions are included in the initial LHS design for both years.

ID	Parameter	2016 bounds	2021 bounds	Years
P1	R-min 660-1000	[0.30, 0.75]	[0.30, 0.75]	Both
P2	R-time 660-1000	[4.00, 16.00]	[4.00, 8.00]	Both
P3	R-min 300-660	[0.30, 0.75]	[0.30, 0.75]	Both
P4	R-time 300-660	[4.00, 16.00]	[4.00, 8.00]	Both
P5	R-min 0-300	[0.30, 0.75]	[0.30, 0.75]	Both
P6	R-time 0-300	[4.00, 12.00]	[2.00, 6.00]	Both
P7	C-min 660-1000	[0.30, 0.75]	[0.30, 0.75]	Both
P8	C-time 660-1000	[4.00, 16.00]	[4.00, 16.00]	Both
P9	C-min 300-660	[0.30, 0.75]	[0.30, 0.75]	Both
P10	C-time 300-660	[4.00, 16.00]	[4.00, 16.00]	Both
P11	C-min 0-300	[0.30, 0.75]	[0.30, 0.75]	Both
P12	C-time 0-300	[4.00, 12.00]	[4.00, 12.00]	Both
P13	MLT band	[0.05, 0.40]	[0.05, 0.40]	Both

Retrofitted coal is a policy-defined group, whereas CHP and its seasonal constraints are technology-defined. Legacy exploration files use `CHP_Retrofitted_*` as an alias for `Coal_Retrofitted_*`; the alias does not imply that all retrofitted units are CHP. Only the technology-defined CHP subset receives the seasonal multiplier in S1.

Table S1 reports the year-specific bounds used in the LHS designs. The 100 evaluations in each year are partitioned into an 80-point development set and an independent 20-point held-out test set. The held-out test set are excluded from GP fitting, cross-validation, and ARD screening.

S3 GP Training, Validation, and ARD Screening

S3.1 Specification and validation

The response is the same quantity mismatch as in the main paper:

$$L(\theta) = \frac{1}{4} \left(E_{\text{coal}}^2 + E_{\text{wind}}^2 + E_{\text{solar}}^2 + E_{\text{exchange}}^2 \right). \quad (\text{S18})$$

Generation errors use capacity-normalized regional RMSE, averaged with equal regional weights; exchange uses the observed-flow normalization described in the main paper. No square root is taken after the weighted squared component sum. No Box–Cox transformation is applied in either year. Inputs and loss are centered and scaled using training data, with fold-specific preprocessing during cross-validation.

The reported models use zero-mean GPs with ARD Matérn kernels. The reported 2016 model uses Matérn 5/2, whereas the reported 2021 model uses Matérn 3/2:

$$\begin{aligned} k_{3/2}(z, z') &= \sigma_f^2 (1 + \sqrt{3}r) \exp(-\sqrt{3}r), \\ k_{5/2}(z, z') &= \sigma_f^2 (1 + \sqrt{5}r + 5r^2/3) \exp(-\sqrt{5}r), \\ r^2 &= \sum_j (z_j - z'_j)^2 / \ell_j^2. \end{aligned} \quad (\text{S19})$$

A single fitted nugget σ_n^2 enters the covariance diagonal. It accommodates numerical tolerances and unresolved response variation; it is not an estimated uncertainty distribution over effective operational parameters. The kernel family is held fixed in the reported ARD and reduced-GP fits.

Table S2: GP configuration used for the reported fits. Bounds are in natural logs.

Item	Setting
Kernel and mean	Matérn 5/2 ARD (2016) and Matérn 3/2 ARD (2021); zero mean after standardization.
Hyperparameter estimation	Bounded maximum marginal likelihood; no hyperpriors.
Bounds	$\log \ell_j \in [-3, 4]$; $\log \sigma_f \in [-3, 3]$; $\log \sigma_n \in [\log(10^{-3}), \log(0.5)]$.
Three initializations	$\ell_j \in \{0.5, 1, 2\}$, with $\sigma_f = 1$, $\sigma_n = 0.05$.
Validation	Independent 20-point held-out test set; five development folds.
ARD criterion	Drop when $\log \ell_j \geq 3.9$ in at least four of five fits.

Table S3: Full- and reduced-GP validation. Predictions are evaluated on the fixed independent held-out test set.

Year	GP	CV R^2	CV RMSE ^a	Test R^2	Test RMSE ^b	n_{test}
2016	Full	0.9982	0.0424	0.9979	1.387	20
2016	Reduced	0.9988	0.0343	0.9987	1.084	20
2021	Full	0.9790	0.1440	0.9936	1.389	20
2021	Reduced	0.9853	0.1204	0.9933	1.419	20

^aCV RMSE uses the saved standardized responses; pooled CV R^2 is reported, not the mean of fold R^2 values. ^bTest RMSE is on the original loss scale, in units of 10^{-4} . Each held-out test set contains 20 observations.

The held-out data serve reporting rather than kernel or feature selection. Test R^2 uses the held-out mean as the constant reference. The reduced models achieve test $R^2 = 0.9987$ in 2016 and 0.9933 in 2021, with original-scale RMSE of 1.08×10^{-4} and 1.42×10^{-4} , respectively. Relative to the corresponding full fits, the reduced fit improves the 2016 test metrics slightly and changes the 2021 metrics only marginally: test R^2 changes from 0.9936 to 0.9933 and RMSE from 1.39×10^{-4} to 1.42×10^{-4} . The reduced-GP 95% predictive-interval coverage is 90% for 2016 and 100% for 2021.

S3.2 Fold-wise ARD screening criterion

For each of the five development folds, the full 13-dimensional GP is refitted on the remaining four folds using fold-specific preprocessing and bounded maximum-marginal-likelihood estimation. Let $\hat{\ell}_j^{(f)}$ denote the fitted ARD length scale for parameter j in fold f .

Because the inputs are standardized, a length scale driven to the imposed upper bound indicates that the fitted loss surface resolves little variation along that parameter over the sampled domain. We therefore define

$$v_j = \sum_{f=1}^5 \mathbf{1} \left\{ \log \hat{\ell}_j^{(f)} \geq 4 - 0.1 \right\}. \quad (\text{S20})$$

Parameter j is screened out when

$$v_j \geq 4, \quad (\text{S21})$$

i.e., when its fitted length scale reaches the upper-bound neighborhood in at least four of the five folds. Otherwise, the parameter is retained for the subsequent reduced-space analysis.

This rule is deliberately conservative: failure to screen out a parameter indicates only that the available design does not provide sufficiently consistent evidence of a flat loss direction. It does not establish that the parameter is precisely recoverable. Parameter-level recoverability is assessed separately using the low-mismatch-region diagnostics in Section S4. Table S4 reports the fold-specific log length scales, upper-bound hits, and resulting retain/screen-out decision for all 13 candidate dimensions.

Table S4: Full 13-dimensional ARD screening. F1–F5 are fitted log length scales; hits are counted before rounding. R/C labels follow Table S1.

Year	Parameter	F1	F2	F3	F4	F5	Hits	Rule
2016	R-min 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-min 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-min 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	R-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-min 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	C-min 300-660	2.18	2.06	2.00	2.07	2.06	0/5	Retain
2016	C-time 300-660	4.00	4.00	4.00	3.85	4.00	4/5	Drop
2016	C-min 0-300	2.07	2.11	1.99	2.06	2.17	0/5	Retain
2016	C-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2016	MLT band	1.29	1.40	1.26	1.31	1.23	0/5	Retain
2021	R-min 660-1000	4.00	4.00	4.00	3.41	4.00	4/5	Drop
2021	R-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	R-min 300-660	2.52	2.67	2.49	2.37	2.39	0/5	Retain
2021	R-time 300-660	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	R-min 0-300	2.73	4.00	2.74	4.00	4.00	3/5	Retain
2021	R-time 0-300	4.00	3.13	2.86	4.00	3.02	2/5	Retain
2021	C-min 660-1000	4.00	4.00	4.00	3.92	4.00	5/5	Drop
2021	C-time 660-1000	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	C-min 300-660	2.03	2.51	2.49	1.65	2.14	0/5	Retain
2021	C-time 300-660	4.00	4.00	3.62	4.00	4.00	4/5	Drop
2021	C-min 0-300	2.34	2.49	2.50	2.45	2.51	0/5	Retain
2021	C-time 0-300	4.00	4.00	4.00	4.00	4.00	5/5	Drop
2021	MLT band	0.84	1.06	0.99	1.09	1.16	0/5	Retain

S3.3 Reduced GP and retained dimensions

Under the baseline ARD criterion, the 2016 baseline screening retains three parameters: MLT band and non-retrofitted CHP minimum output in the 300–660 and 0–300 MW groups. Under the same

criterion, the 2021 screening retains these three plus retrofitted-coal minimum output in the 300–660 and 0–300 MW groups and retrofitted-coal minimum up/down time in the 0–300 MW group, for six retained dimensions in total.

The reduced GP is refitted using only these retained coordinates from the development design. It approximates the reduced empirical response represented by that design. It is neither the loss conditional on a particular value of the discarded parameters nor an exact analytical marginalization over them. Variation in omitted dimensions can contribute to residual variation.

The 2021 R-min 0–300 dimension reaches the ceiling in three folds and the R-time 0–300 dimension in two folds. Both therefore remain below the specified four-of-five exclusion threshold. A rounded median log length scale alone would not reproduce these fold-wise decisions.

Screening stability and cross-fitted validation

Sensitivity to the length-scale ceiling is assessed by repeating the five-fold screening with *length-scale* ceilings multiplied by 0.5, 1, and 2: log ceilings $4 + \log(0.5)$, 4, and $4 + \log 2$. The hit tolerance remains 0.1 log units.

Table S5: Ceiling hits under alternative upper bounds. Four or more hits imply exclusion under that setting.

Parameter	2016: 0.5×	1×	2×	2021: 0.5×	1×	2×
R-min 660-1000	5/5	5/5	5/5	5/5	4/5	4/5
R-time 660-1000	5/5	5/5	5/5	5/5	5/5	5/5
R-min 300-660	5/5	5/5	5/5	0/5	0/5	0/5
R-time 300-660	5/5	5/5	5/5	5/5	5/5	5/5
R-min 0-300	5/5	5/5	5/5	3/5	3/5	3/5
R-time 0-300	5/5	5/5	5/5	3/5	2/5	2/5
C-min 660-1000	5/5	5/5	5/5	5/5	5/5	4/5
C-time 660-1000	5/5	5/5	3/5	5/5	5/5	5/5
C-min 300-660	0/5	0/5	0/5	0/5	0/5	0/5
C-time 300-660	5/5	4/5	5/5	5/5	4/5	4/5
C-min 0-300	0/5	0/5	0/5	0/5	0/5	0/5
C-time 0-300	5/5	5/5	5/5	5/5	5/5	4/5
MLT band	0/5	0/5	0/5	0/5	0/5	0/5

The 2021 retain/drop set is unchanged across the three tested ceilings. In 2016, doubling the ceiling changes C-time 660–1000 from excluded to retained: its ceiling-hit count falls from five to three. The three parameters retained by the 2016 baseline remain retained under every tested ceiling, but the complete 2016 retain/drop set is therefore not invariant to the upper-bound specification.

A five-outer-fold check repeats five-fold ARD inside each outer training set and compares full/reduced predictions on the excluded fold, addressing selection leakage in fixed-active-set CV.

Table S6: Cross-fitted full/reduced comparison. RMSE is in standardized response units; the change is relative to the full model.

Year	Full R^2	Reduced R^2	Full RMSE	Reduced RMSE	Δ RMSE
2016	0.9981	0.9988	0.0433	0.0343	-20.74%
2021	0.9790	0.9824	0.1440	0.1321	-8.26%

The cross-fitted reduced models improve aggregate out-of-fold prediction: RMSE decreases by 20.74% in 2016 and 8.26% in 2021. The corresponding R^2 values change from 0.9981 to 0.9988 and from 0.9790 to 0.9824.

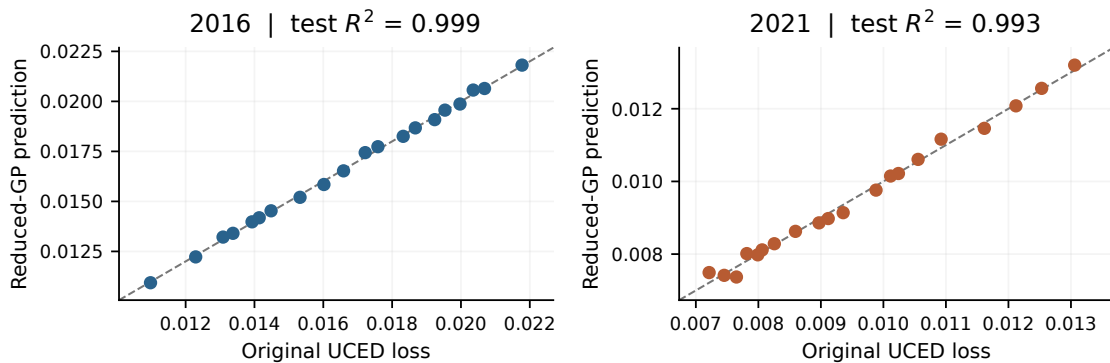


Figure S1: Reduced-GP predictions on independent held-out evaluations. Dashed lines indicate equality; All held-out test set are shown.

S4 Low-Mismatch Region and Recoverability Diagnostics

S4.1 Definition and primary region

For each year, the reduced GP is evaluated on a fixed 20,000-point stratified LHS pool in the retained-parameter space (seed 123). The coordinates span the domain in S2; no original UCED solves are added. With $\hat{L}_{min}$ denoting the minimum predicted loss in that pool,

$$S_\epsilon = \{\theta_A \text{ in the evaluated pool} : \hat{L}(\theta_A) \leq (1 + \epsilon)\hat{L}_{min}\}, \quad \epsilon = 0.10. \quad (\text{S22})$$

This finite-pool approximation describes a relative low-mismatch region, not a confidence or credible region. Its reference minimum is not asserted to be a global optimum. For dimension j , concentration is summarized by

$$\text{nIQR}_j(S_\epsilon) = \frac{Q_{0.75}(\theta_j | S_\epsilon) - Q_{0.25}(\theta_j | S_\epsilon)}{U_j - L_j}. \quad (\text{S23})$$

$[L_j, U_j]$ is the corresponding design domain. Lower nIQR indicates greater concentration within this pool and tolerance, not unique identification.

The minimum predicted losses in the evaluated pools are 0.0104338 in 2016 and 0.0071058 in 2021. The primary regions contain 260 (1.30%) and 3420 (17.10%) of their respective pools.

Table S7: Retained-parameter distributions within the primary $S_{10\%}$ region. Minimum-output fractions and the MLT band are dimensionless; R-time is reported in hours. Min/max are finite-pool extrema.

Year	Parameter	Min	Q25	Median	Q75	Max	nIQR
2016	C-min 300-660	0.501	0.633	0.677	0.719	0.749	0.192
2016	C-min 0-300	0.477	0.621	0.669	0.714	0.748	0.208
2016	MLT band	0.050	0.054	0.060	0.068	0.095	0.039
2021	R-min 300-660	0.300	0.446	0.559	0.659	0.750	0.473
2021	R-min 0-300	0.300	0.422	0.536	0.645	0.750	0.495
2021	R-time 0-300	2.001	3.177	4.198	5.112	5.997	0.484
2021	C-min 300-660	0.301	0.436	0.528	0.648	0.750	0.473
2021	C-min 0-300	0.300	0.416	0.530	0.639	0.750	0.495
2021	MLT band	0.050	0.065	0.080	0.098	0.167	0.095

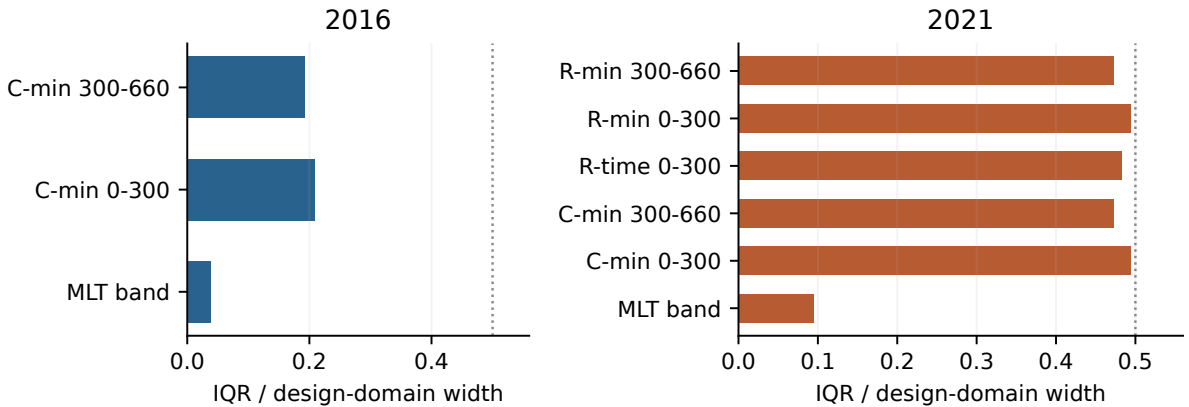


Figure S2: Domain-normalized dispersion in $S_{10\%}$. The dotted 0.5 reference is the IQR of a uniform distribution over the full domain, not a statistical decision threshold.

MLT is more concentrated than the retained coal minimum-output parameters in both years (nIQR 0.039 and 0.095). The additional 2021 retrofitted-coal dimensions have nIQR near 0.5: their ARD relevance does not translate into narrow parameter-level ranges.

S4.2 Tolerance sensitivity and joint relationships

Table S8: Normalized IQR under alternative relative mismatch tolerances.

Year	Parameter	5%	10%	20%	30%
2016	C-min 300-660	0.121	0.192	0.395	0.472
2016	C-min 0-300	0.123	0.208	0.378	0.472
2016	MLT band	0.020	0.039	0.065	0.094
2021	R-min 300-660	0.330	0.473	0.491	0.497
2021	R-min 0-300	0.495	0.495	0.501	0.501
2021	R-time 0-300	0.476	0.484	0.497	0.499
2021	C-min 300-660	0.442	0.473	0.488	0.486
2021	C-min 0-300	0.474	0.495	0.503	0.500
2021	MLT band	0.047	0.095	0.177	0.242

Table S9: Number of pool members in each tolerance region; each pool contains 20,000 points.

Year	5%	10%	20%	30%
2016	53	260	1569	3415
2021	859	3420	7041	9676

The MLT band remains the most concentrated retained dimension at each tested tolerance in both years. Its range broadens as the tolerance increases, especially in 2021. Coal-parameter dispersion is also tolerance-dependent; the narrowest 2016 region has relatively few sampled members and should not be read as a precise interval estimate. All summaries remain conditional on the parameter bounds and the fitted reduced surface.

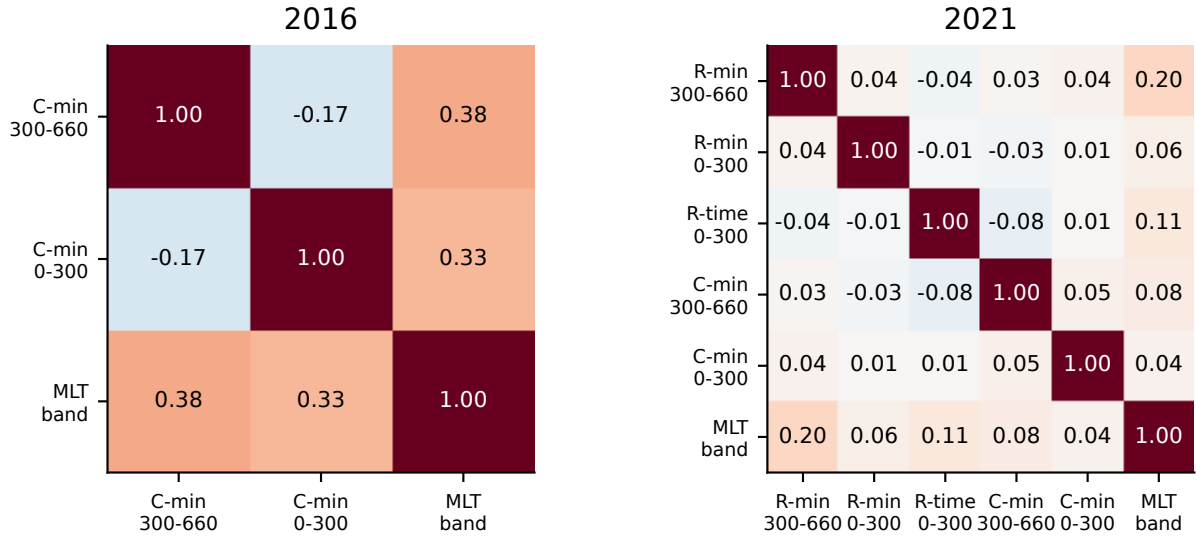


Figure S3: Spearman rank correlations among retained parameters within the $S_{10\%}$. These describe the selected region's geometry; they are neither causal effects nor tests of structural identification.

The two non-retrofitted CHP minimum-output dimensions are negatively associated in the 2016 region ($\rho = -0.17$, ρ denotes Spearman rank correlation with the equal-weight ordering.), whereas the corresponding association is weak in 2021. Relationships with MLT can reflect compensation between delivery flexibility and minimum-generation rigidity, but pairwise association alone does not establish a unique mechanism. The weak 2021 coal-coal correlations also illustrate that broad parameter ranges need not follow a single narrow tradeoff curve.

Interpretation and scope of the region diagnostics

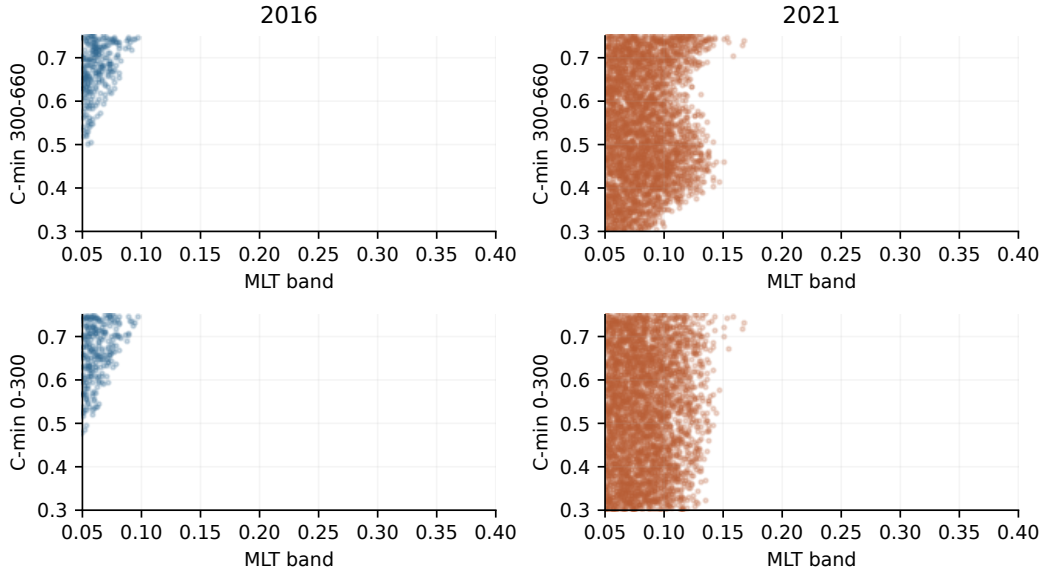


Figure S4: Pairwise projections of $S_{10\%}$ for MLT and the two shared non-retrofitted CHP minimum-output parameters. All region members are plotted; projections conceal variation in other retained dimensions.

The evidence supports different levels of interpretation across parameters. MLT has a relatively concentrated model-ready delivery representation; the coal minimum-output dimensions support a broader diagnosis of operating rigidity. None of these sets uniquely identifies a policy or a causal reform effect. Cross-year differences are interpreted as regime-specific effective UCED representations, conditional on each year’s inputs, observations, and parameter domain.

The lower MLT domain boundary is 0.05. Concentration near this edge cannot show whether a still narrower band would also match the quantities, because that region was not evaluated. The 2016 screening sensitivity in S2 and the limited spatial coverage of the six-point-per-year revalidation in S5 constrain how specifically the surrogate region can be interpreted. In particular, GP–UCED discrepancies at fixed values of omitted parameters can combine approximation error and variation associated with those parameters; they are not a formal upper bound on either component.

S5 Alternative Target Weights

The baseline loss assigns equal weight to coal generation, wind generation, solar generation, and interregional exchange. To isolate sensitivity to this aggregation choice from changes in the evaluated design, the same 100 original UCED evaluations in each year are re-ranked under

$$L_w(\theta) = w_c E_{coal}^2 + w_w E_{wind}^2 + w_s E_{solar}^2 + w_m E_{exchange}^2, \quad \sum_k w_k = 1. \quad (\text{S24})$$

Besides the equal baseline (1/4, 1/4, 1/4, 1/4), the alternatives are coal-heavy (1/2, 1/6, 1/6, 1/6), MLT-heavy (1/6, 1/6, 1/6, 1/2), and excluding-MLT (1/3, 1/3, 1/3, 0). The first two perturb relative emphasis while retaining every target. The last is a leave-one-target-out diagnostic, not a preferred objective.

Table S10 reports Spearman rank correlation with the equal-weight ordering, overlap with its ten lowest-loss evaluations, and the best observed design point. The final column gives the 25th percentile, median, and 75th percentile of the MLT band among the ten lowest-loss points. Loss magnitudes should not be compared across rows with different weights.

Table S10: Sensitivity to alternative target weights using the same 100 original UCED evaluations per year. The overlap column counts members shared with the equal-weight top ten; Top-10 MLT is reported as Q25/median/Q75.

Year	Scheme	ρ	Overlap	Best run	Best loss	Best MLT	Top-10 MLT
2016	Equal	1.0000	10/10	61	0.010980	0.050	0.060/0.067/0.075
2016	Coal-heavy	0.9996	10/10	61	0.010047	0.050	0.060/0.067/0.075
2016	MLT-heavy	0.9947	10/10	61	0.008975	0.050	0.060/0.067/0.075
2016	Excluding MLT	0.8418	5/10	13	0.012917	0.101	0.073/0.121/0.160
2021	Equal	1.0000	10/10	6	0.007212	0.075	0.060/0.067/0.075
2021	Coal-heavy	0.9959	9/10	6	0.010035	0.075	0.060/0.070/0.082
2021	MLT-heavy	0.9991	10/10	6	0.006542	0.075	0.060/0.067/0.075
2021	Excluding MLT	0.8224	5/10	88	0.007813	0.094	0.066/0.086/0.105

Coal-heavy and MLT-heavy weighting leave the best observed run unchanged in both years. Their rank correlations with the equal-weight loss are at least 0.995, and nine or ten of the equal-weight top ten remain in the reweighted top ten. The MLT-band quartiles are unchanged under MLT-heavy weighting and change only slightly under coal-heavy weighting in 2021. Thus the ordering of the evaluated designs is insensitive to these moderate reallocations among included targets.

Excluding the MLT target produces a larger change in the ranking: rank correlation falls to 0.842 in 2016 and 0.822 in 2021, and top-ten overlap falls to five in both years. The top-ten MLT median shifts from 0.067 under equal weighting to 0.121 in 2016 and 0.085 in 2021, with a wider interquartile range, especially in 2016. The concentration of the MLT band is substantially weakened when the exchange target is excluded; it therefore should not be interpreted as recovery from generation quantities alone.

S6 Original UCED Revalidation and Quantity Matching

For each year, six parameter vectors were evaluated using the original UCED: a center representative, four interior points, and a surrogate-best point. The six points were selected from the low-mismatch pool using the criteria described below, before original UCED revalidation. GP predictions are evaluated at the rounded parameter vectors actually supplied to the UCED, so GP and UCED values are compared at identical retained-parameter coordinates.

S6.1 2016 revalidation

The center representative is nearest the region centroid in domain-normalized coordinates, and the four interior points span distance quartiles within the region. The surrogate-best point is the lowest-predicted-loss pool member. All six satisfy the 2016 $S_{10\%}$ cutoff, approximately 0.011477. Omitted parameters remain fixed at their prescribed midpoint values.

Table S11: 2016 original UCED revalidation. The GP–UCED discrepancy is defined as UCED loss minus GP-predicted loss.

Point	GP loss	UCED loss	UCED – GP
Center representative	0.011201	0.011088	-0.000112
Interior 1	0.010955	0.010754	-0.000201
Interior 2	0.011148	0.010922	-0.000225
Interior 3	0.011286	0.011151	-0.000136
Interior 4	0.011442	0.011189	-0.000253
Surrogate best	0.010423	0.010087	-0.000335

The six UCED losses range from 0.010087 to 0.011189. The reduced GP predicts higher losses than the midpoint-completed UCED evaluations at all six points, with a mean absolute discrepancy of 0.000210 and a maximum of 0.000335. The median absolute relative discrepancy is 1.97%. The surrogate-best point has the lowest UCED loss among these six; the center and four interior points range from 0.010754 to 0.011189. Thus the evaluations provide evidence of comparable matching at several selected representations, while also revealing local GP–UCED discrepancies.

Table S12: 2016 component NRMSE values. Aggregate loss is one quarter of the sum of the four squared components, using unrounded values.

Point	Coal	Wind	Solar	Exchange
Center representative	0.0840	0.1297	0.1214	0.0757
Interior 1	0.0832	0.1275	0.1193	0.0750
Interior 2	0.0838	0.1274	0.1190	0.0792
Interior 3	0.0844	0.1304	0.1221	0.0746
Interior 4	0.0838	0.1321	0.1231	0.0715
Surrogate best	0.0816	0.1242	0.1167	0.0681

Coal NRMSE ranges from 0.0816 to 0.0844, wind from 0.1242 to 0.1321, solar from 0.1167 to 0.1231, and exchange from 0.0681 to 0.0792. The aggregate agreement should therefore be read together with component errors rather than as evidence of equally close matching for every quantity.

S6.2 2021 revalidation

The 2021 center representative is nearest the region centroid in domain-normalized coordinates, and the four interior points are sampled from distance quartiles. The surrogate-best point is the lowest-predicted-loss pool point and is included as a low-predicted-loss reference point. All six satisfy its 2021 $S_{10\%}$ loss cutoff, approximately 0.007816. Omitted parameters remain fixed at their prescribed midpoint values.

Table S13: 2021 original UCED revalidation. The GP–UCED discrepancy is defined as UCED loss minus GP-predicted loss. Point selection is based on the continuous pool coordinates before parameters are rounded for UCED evaluation.

Point	GP loss	UCED loss	UCED – GP
Center representative	0.007503	0.007161	-0.000342
Interior 1	0.007702	0.007371	-0.000331
Interior 2	0.007335	0.007168	-0.000168
Interior 3	0.007767	0.007309	-0.000458
Interior 4	0.007684	0.007624	-0.000059
Surrogate best	0.007107	0.007065	-0.000042

The six original UCED losses range from 0.007065 to 0.007624. The GP prediction exceeds the corresponding midpoint-completed UCED loss at all six points, with a mean absolute residual of 0.000233 and a maximum of 0.000458. The median absolute relative discrepancy is 3.41%. These are descriptive checks at selected points, not an independent estimate of region-wide prediction accuracy. The surrogate-best point yields the lowest evaluated loss among these six, 0.007065, and its residual magnitude is about 0.59% of its GP prediction. Several distinct parameter vectors retain similar quantity mismatch under the original model.

Table S14: 2021 component NRMSE values. The aggregate loss equals one quarter of the sum of their squares, using unrounded values.

Point	Coal	Wind	Solar	Exchange
Center representative	0.1232	0.0828	0.0362	0.0727
Interior 1	0.1230	0.0829	0.0360	0.0787
Interior 2	0.1241	0.0830	0.0362	0.0712
Interior 3	0.1229	0.0829	0.0362	0.0771
Interior 4	0.1255	0.0830	0.0350	0.0815
Surrogate best	0.1233	0.0829	0.0364	0.0697

Coal, wind, and solar errors vary little across these points; exchange errors show a wider range (0.0697–0.0815). The component results support similar aggregate matching at the selected representations, without establishing uniform accuracy across all hours, regions, or all of $S_{10\%}$. The checks support the use of a low-mismatch region rather than identifying one uniquely preferred institutional parameterization. They do not resolve the screening sensitivity or parameter-domain limitations discussed above.

S7 Reproducibility and Repository Contents

The accompanying repository contains tables, figures, plotting data, and a SHA-256 source manifest. Scripts regenerate held-out and dense-pool predictions from fitted GP model objects. The repository manifest records Julia 1.11.3 and GaussianProcesses.jl 0.12.6. Fixed train/test splits, cross-validation folds, parameter domains, and fitted-model files preserve the run-specific settings.